ScholAR: Page-Grounded Retrieval-Augmented Generation for Scientific Document Comprehension

Anonymous submission

Abstract

Understanding a scientific paper means being able to trace each claim back to the exact place it came from. General-purpose chat over an uploaded PDF is fluent but unreliable for this: nothing ties a cited page number to the text the model was actually shown, and the scientific retrieval-augmented systems that do better tend to rely on cloud-hosted frontier models or datastores of millions of papers. We present ScholAR (Smart Companion for Holistic Organization, Literature Analysis & Research), a retrieval-augmented generation system for scientific documents that runs entirely on a consumer laptop through a local model server, with no cloud API for text or vision. Two design choices carry the system. Page-preserving chunking keeps every retrieved unit inside a single PDF page, so a citation resolves to one highlightable location. Indirect citation grounding lets the model cite only evidence identifiers that map to chunks it actually retrieved, so it cannot write a page number of its own. We evaluate retrieval, answer faithfulness, visual grounding, and cross-document localization on labeled benchmarks. On 51 cases, the generated answers stay grounded in their retrieved context with a mean score of 0.971, no answer atom contradicts its source, and 94% of inline citations are fully supported with none unsupported. BM25-primary retrieval reaches Recall@5 of 0.93 across 25 papers, and figure and table questions route to the correct visual element in all 18 pilot cases. Cross-document localization sits at or below a random-guess floor, which we report as our clearest open problem. Against local reimplementations of PDF-chat, vanilla RAG, and a PaperQA2-style pipeline, ScholAR leads on answer correctness and faithfulness, and when a question's answer is absent from the retrieved paper the local models decline rather than fabricate on 90 to 100% of a provably-unanswerable set, each within an 18 GB single-machine budget. We also contribute a 100-case benchmark across 25 papers spanning text, mathematical, and figure questions, and a model-agnostic human-evaluation protocol over four local models whose expert scoring is pending.

1 Introduction

Reading an unfamiliar paper is slowest at the moments that matter most: checking that a stated number, a method detail, or a figure actually says what you think it says. Uploading the PDF to a general-purpose chat model helps with fluency but brings a new failure mode. Such systems summarize confidently while inventing page references and attributing quotes to places they do not appear, because a generic chat pipeline puts no hard constraint between a citation and the text the model was conditioned on.

Purpose-built scientific retrieval-augmented generation (RAG) systems close part of this gap, but they assume resources most readers do not have on their own machine. OpenScholar (Asai et al. 2024), PaperQA2 (Skarlinski et al. 2024), and SciRAG (Ding et al. 2025) depend on cloud-hosted frontier language models, and OpenScholar additionally on a datastore of tens of millions of papers. We ask a narrower question: what does grounded, citable question answering over a scientific document look like under a hard local-only constraint, with a single sub-15-billion-parameter model, no cloud API, and a retrieval design simple enough to audit end to end?

Two ideas carry ScholAR. Page-preserving chunking segments a document so that every retrieved chunk lies entirely within one PDF page, which makes a citation resolve to an exact, highlightable location rather than an approximate span. Indirect citation grounding assigns each retrieved passage a short-lived evidence identifier and instructs the model to cite only those identifiers, never a free-form page number; the application then rewrites each identifier into a numbered reference that maps deterministically to a page. The model decides only which evidence supports a claim, and the evidence-to-page mapping is application logic rather than a model guess. Beyond single-document text, ScholAR routes figure and table questions to a locally run, natively multimodal model and extends to a paper's own cited references for cross-document questions.

Our contributions are:

- A fully local RAG system for scientific documents that runs both text and vision inference on-device, using page-preserving chunking and an indirect citation mechanism that rules out page-number hallucination by construction.
- An evaluation that scores the faithfulness of the system's *generated answers* rather than of pre-written gold claims, paired with an automated per-citation support metric. On 51 cases, generated answers reach a mean grounding score of 0.971 with a zero contradiction rate and 94% fully supported citations.

- Automated benchmarks and results for retrieval, retrieval-support faithfulness, visual grounding, and cross-document localization, each reported with its limitations, including a cross-document result at or below a random-guess floor.
- A 100-case benchmark across 25 papers spanning text, mathematical, and figure questions, with gold answers verified against their source passages, and a model-agnostic human-evaluation protocol over four local models. The expert scoring is under way, and its result tables are marked as placeholders here.

2 Related Work

Retrieval-augmented generation. The RAG framework couples a retriever with a generator so that outputs are conditioned on retrieved evidence (Lewis et al. 2020). Lexical retrieval via BM25 (Robertson and Zaragoza 2009) remains a strong baseline, and the BEIR benchmark (Thakur et al. 2021) documents that zero-shot dense retrievers often fail to beat it. ScholAR uses BM25 as its primary retrieval signal and dense retrieval as a fusion component.

Scientific-literature RAG systems. PaperQA and PaperQA2 (Lála et al. 2023; Skarlinski et al. 2024) are agentic systems that report high accuracy on scientific literature question answering using frontier model backends. OpenScholar (Asai et al. 2024) synthesizes literature over a large open datastore, and SciRAG (Ding et al. 2025) adds citation-graph-aware, outline-guided synthesis. All three assume cloud-hosted or large self-hosted models. ScholAR differs by targeting a single local sub-15-billion-parameter model on consumer hardware, trading raw capability for zero data exposure and no API dependency.

Small models that cite their sources. Closest in spirit to our citation mechanism, Pleias-RAG (Pleias 2025) trains small (350M and 1.2B) models that emit literal source quotes during generation. That line shows citation behavior can be learned into a compact model. ScholAR takes a complementary, model-agnostic route: the grounding lives in the retrieval and application layer through evidence identifiers, so it works with any local generation model rather than requiring a model trained to quote.

Citation and faithfulness evaluation. Gao et al. (2023) formalize citation precision and recall for generated text using entailment between cited passages and statements. SummaC (Laban et al. 2022) scores summary consistency with a zero-shot, sentence-level entailment variant that needs no fine-tuned model, and FActScore (Min et al. 2023) decomposes text into atomic facts verified against a source. RAGAS (Es et al. 2024) automates RAG faithfulness scoring against retrieved context. Our faithfulness metric adopts the SummaC-style zero-shot entailment and the FActScore-style atomic decomposition, and we apply it to the system's own generated answers.

Visual and multi-document scientific QA. ColPali (Faysse et al. 2024) indexes pages as visual embeddings for retrieval without OCR. ScholAR keeps text retrieval primary and routes only figure and table questions to a cropped-region vision call, which keeps the common text path fast and fully lexical. M3SciQA (Li et al. 2024) pairs each anchor paper with its cited references and finds that even large foundation models underperform human experts at cross-paper reasoning, a difficulty our cross-document results echo. Document parsing tools such as GROBID (Lopez 2009) and Nougat (Blecher et al. 2023) target structured extraction and are complementary to our chunking.

3 Problem Formulation

Given a scientific PDF D, a user question q, and a set of extracted chunks $\mathcal{C} = \{c_1, \ldots, c_n\}$ derived from D (and, in multi-document mode, from D's cited references), the system retrieves a ranked subset $\mathcal{C}_q \subseteq \mathcal{C}$ and generates an answer $\hat{a}$ grounded in that evidence:

$$\hat{a} = \arg\max_{a} P(a \mid q, \mathcal{C}_q), \quad \mathcal{C}_q = \mathrm{Retrieve}(q, \mathcal{C}, k). \quad (1)$$

Each chunk c_i carries a source page number and, in multi-document mode, a source paper identifier, so $\mathcal{C}_q$ always resolves to a specific page of a specific document. Generation attaches each claim in $\hat{a}$ to an evidence identifier referencing a specific $c_i \in \mathcal{C}_q$ rather than a free-form page number, so citations are constrained to what was actually retrieved, independent of $\mathrm{Retrieve}$'s own precision.

4 Method

Figure 1 shows the end-to-end pipeline. A PDF is chunked one page at a time, ranked by BM25, and exposed to the model as evidence identifiers; the model's identifier citations are then rewritten into page-anchored references by the application rather than by the model.

4.1 Document Processing and Page-Preserving Chunking

Given an arXiv identifier or an uploaded PDF, ScholAR extracts per-page text with PyMuPDF (Artifex Software 2024) and persists metadata, pages, chunks, and figures as JSON. Standard character or token windows routinely split a chunk across a page boundary, breaking the one-to-one map between a retrieved chunk and the page a citation should highlight. ScholAR instead chunks strictly within each page: it slides a window of target size 1400 words with 120-word overlap over each page's tokens independently, so a chunk never spans two pages. A lightweight regular-expression pass tags each chunk with a detected section title when a heading is found near the top of the page. Chunk identifiers are numbered per paper.

4.2 BM25-Primary Retrieval

The production retriever ranks chunks with Okapi BM25 (Robertson and Zaragoza 2009) (k_1=1.4, b=0.72) as the primary signal, then applies small additive heuristic boosts: a query-expansion dictionary maps research terms to related vocabulary, an explicit page number in the query adds a page-hint boost, and query language implying a section or a claim-bearing phrase adds a small boost. A separate

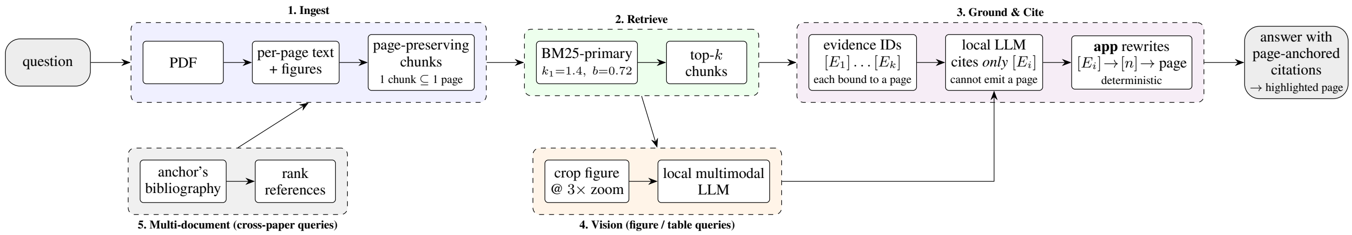

Figure 1: The ScholAR architecture. A question enters at the left and leaves as an answer whose every citation resolves to a highlightable page. **(1)** Ingest keeps each chunk inside a single PDF page. **(2)** BM25-primary retrieval returns the top-k chunks. **(3)** Those chunks are exposed to the model only as evidence identifiers $[E_i]$, each already bound to the page of the chunk it came from; the model may cite an identifier but can never write a page number, and the *application*, not the model, rewrites $[E_i]$ into a numbered reference and resolves it to a page. A cited page is therefore correct by construction rather than by the model's good behavior. **(4)** Figure and table queries additionally route a cropped, 3×-zoomed image to the same local multimodal model. **(5)** Cross-paper queries first rank the anchor's bibliography, and the selected reference is then ingested like any other paper. Every component runs on one machine.

hybrid configuration fuses BM25 with dense retrieval over a local all-MiniLM-L6-v2 sentence encoder (Reimers and Gurevych 2019) using Reciprocal Rank Fusion (Cormack, Clarke, and Buettcher 2009) (k=60); this configuration is used in the faithfulness study.

4.3 Indirect Citation Grounding

Asking a model to write a page number directly (for example "[p. 7]") is fragile, because the model has no reliable way to know which page a claim came from and will produce a plausible-looking one. ScholAR instead assigns each retrieved passage a short-lived evidence identifier for the current turn and instructs the model to cite only these identifiers. After generation, the system discards any page number the model wrote on its own, rewrites each valid evidence identifier into a compact numbered reference in citation order, and resolves each reference to its source chunk's page for highlighting. The model decides only which evidence supports a claim; the mapping from evidence to page is deterministic application logic. This prevents page-number hallucination by construction rather than by asking the model to be careful.

4.4 Visual Grounding

When a query references visual content, through cue terms such as "figure", "table", or an explicit figure number, retrieval boosts figure and table chunks so they can compete against longer text chunks. Each caption is detected by regular expression, and the region above it is rendered at 3× zoom and cached alongside the caption. A query naming an explicit figure number is matched directly against that figure's label, which matters because a letter-initial tokenizer reduces "Figure 2" and "Figure 14" to the same token once digits are dropped. When the top-ranked chunk is a figure or table, the cached image is sent, with the caption and supporting text, to the same local model used for text generation (both models we use are natively multimodal), rather than to a separate vision model or cloud API. If the image is missing or the call fails, the system degrades to a caption-only extractive answer.

4.5 Multi-Document Extension

ScholAR resolves an anchor paper's bibliography and lets the user ingest cited works into the same session. For arXiv papers, references are resolved by arXiv identifier; for uploaded PDFs, by a title search over the last pages, matched against numbered and author-year citation styles. Ingesting a reference applies the same page-preserving chunking and tags every chunk with a source paper identifier. At query time, chunks from all loaded papers merge into one retrieval pool, and each citation is labeled with its source paper.

4.6 Faithfulness Metrics

We evaluate faithfulness with a zero-shot, sentence-level entailment approach in the SummaC style (Laban et al. 2022), decomposing text into atomic claims in the FActScore style (Min et al. 2023). For an atom a_i and a chunk $\mathcal{C}_j$ we take the maximum cosine similarity between the atom and the chunk's sentences under the local all-MiniLM-L6-v2 encoder, and label the atom entailment, neutral, or contradiction against thresholds τ_e=0.42 and τ_n=0.25. A Combined Faithfulness Score (CFS) blends three tiers:

$$\mathrm{CFS} = 0.50\,\mathrm{NLI} + 0.30\,\mathrm{SCR} + 0.20\,\mathrm{KFP}, \tag{2}$$

where NLI is the entailment fraction, SCR is whole-claim to best-chunk cosine similarity, and KFP is boundary-matched recall of numbers and technical terms. The thresholds and the CFS$\geq$0.55 faithful cutoff were calibrated on a small pilot set; we did not run a systematic sensitivity sweep.

We apply this metric in two ways that answer different questions. *Retrieval-support CFS* scores a pre-written gold claim against the retrieved chunks, measuring whether retrieval surfaces support for a known fact (answerability). *Generation faithfulness* scores the system's *actual generated answer* against its retrieved context, measuring whether the output stays grounded and avoids contradicting its source. The second is the quantity our hallucination claim rests on, and it is the result we foreground.

4.7 System Architecture

The backend is a local application that stores parsed PDFs as JSON on disk; the frontend highlights cited pages. All inference, both text generation and figure or table question answering, runs locally through a single natively multimodal model, with no cloud API used for inference and no user data leaving the machine. Paper acquisition and reference resolution use public arXiv and Semantic Scholar lookups by identifier only. All experiments in this paper were run on a consumer laptop (Apple Silicon, 18 GB unified memory) under Python 3.12, with all-MiniLM-L6-v2 loaded from a local cache; no discrete or cloud GPU was used, and inference runs on the laptop's integrated GPU through the local model server's accelerated backend.

5 Benchmarks

We use two families of evaluation data. The first is a set of small, manually labeled benchmarks over three landmark NLP papers (Attention Is All You Need (Vaswani et al. 2017), Retrieval-Augmented Generation (Lewis et al. 2020), and LLaMA (Touvron et al. 2023)): a 14-case retrieval benchmark with labeled relevant chunks, a 51-case faithfulness benchmark with oracle claims and a supporting chunk, an 18-case visual-grounding pilot, and an 18-case multi-document benchmark of which 10 have an arXiv-resolvable target. These support the automated metrics that need ground-truth chunk labels, and they are small: the hand-labeled benchmarks cover three papers between them.

The second is a 100-case benchmark spanning 25 papers, built to broaden coverage and reduce the memorization advantage a model has on very famous papers. It contains 50 text, 25 mathematical, and 25 figure or table questions. Each case was mined from a real passage or caption by a local model and then kept only if the required answer facts appear, on a word or number boundary, in the source passage; questions from benchmark-dataset papers, reference lists, and worked examples were filtered out. Because each case carries exact substrings copied from its source passage, we recover a gold supporting chunk for every case at no extra labeling cost: it is the chunk that contains those substrings, disambiguated by the recorded page. This yields a 100-case, 25-paper labeled set for retrieval and retrieval-support faithfulness, an order of magnitude larger and eight times more papers than the hand-labeled set. The labels are weaker than hand annotation, and the questions, being mined from passages, carry lexical overlap that can favor lexical retrieval; we therefore report the scaled set alongside the 3-paper hand-labeled set rather than in place of it. This diverse benchmark is also the dataset for the human evaluation (Section 7). Table 1 summarizes the corpus and the benchmarks built on it.

Table 1: Key statistics of the ScholAR corpus and benchmarks. Page-preserving chunking yields close to one chunk per page, because a typical page holds fewer words than the chunk window.

Corpus		*Diverse benchmark*	
Papers	25	Questions	100
Pages / paper	34.5	text/math/vis.	50/25/25
Chunks / paper	34.8	Question words	12.9
Words / paper	17,694	Gold facts / q.	2.8
Labeled sets (hand / auto)		*Generated answers*	
Retrieval	14 / 100	Answers	350
Faithfulness	51 / 100	Local models	4
Visual / multi-doc	18 / 18	Inline citations	788
Unanswerable	20	Citations / answer	2.3

Table 2: Retrieval by question type on the same 100 scaled cases. Both retrievers degrade from text to mathematical to visual questions; BM25-primary leads dense-only in every category.

	BM25-primary		**Dense-only**	
Question type	**R@5**	**MRR**	**R@5**	**MRR**
Text (n=50)	0.98	0.904	0.82	0.680
Mathematical (n=25)	0.92	0.853	0.76	0.507
Visual (n=25)	0.84	0.781	0.56	0.423

6 Automated Evaluation

6.1 Retrieval

Table 3 reports retrieval on the scaled benchmark of 100 auto-labeled cases across 25 papers. BM25-primary reaches Recall@5 of 0.93 and MRR of 0.861; plain BM25 is within noise at 0.94 and 0.863, so the reranking heuristics and page hints again leave the aggregate unchanged. The clear shift from the 3-paper anchor is dense-only: it led there (Recall@5 1.000, MRR 0.881) but trails every lexical configuration on the diverse pool (MRR 0.572), the expected BEIR outcome (Thakur et al. 2021) once the corpus is not three heavily studied papers. We read the size of BM25's margin cautiously, because the mined queries share lexical surface with their source passages, but the direction, lexical over single-pass dense at scale, is the one the BM25-primary design assumes.

Aggregates hide where retrieval struggles, so Table 2 breaks the same run down by question type. Both retrievers degrade monotonically from text to mathematical to figure and table questions, most steeply on the visual cases (BM25-primary Recall@5 falls from 0.98 to 0.84): a question about a figure shares little surface form with the caption and body text around it, which is why ScholAR routes such questions to a vision model rather than trusting retrieval alone. BM25-primary still leads dense-only in every category, by the widest margin on exactly those visual cases.

6.2 Retrieval-Support Faithfulness

Table 4 reports the retrieval-support CFS on the scaled 100-case set. BM25-primary and the hybrid BM25+Dense+RRF retriever land within 0.003 of each other (Combined CFS 0.785 versus 0.782), and BM25-primary has the higher supporting-chunk hit rate at depth 5 (SCHR@5 0.86 versus 0.75) and one more faithful case (93 versus 92 of 100 under the CFS$\geq$0.55 cutoff). The narrow hybrid edge on the 3-paper anchor (Combined CFS 0.827 versus 0.807,

Table 3: Retrieval on the scaled benchmark: 100 auto-labeled cases across 25 papers, with NDCG@5 over the gold chunks. On the 3-paper hand-labeled anchor (14 cases) dense-only instead leads (R@5 1.000, MRR 0.881).

System	R@1	R@3	R@5	MRR	NDCG@5
Keyword	0.670	0.860	0.950	0.779	0.762
BM25-Only	0.810	0.910	0.940	0.863	0.828
BM25+Rerank	0.810	0.910	0.930	0.861	0.824
Dense-Only	0.470	0.650	0.740	0.572	0.523

Table 4: Retrieval-support faithfulness on the scaled 100-case set (25 papers). NLI is the entailment fraction; SCR is whole-claim cosine; KFP is key-fact recall; SCHR@5 is the supporting-chunk hit rate at depth 5. On the 3-paper anchor (51 oracle-claim cases) hybrid narrowly leads (CFS 0.827 vs 0.807).

Metric	BM25-primary	Hybrid+RRF
NLI (entailment)	0.925	0.915
SCR (cosine)	0.433	0.438
KFP (key-fact recall)	0.966	0.966
Combined CFS	0.785	0.782
SCHR@5	0.860	0.750
Faithful ($\geq$ 0.55)	93/100	92/100

SCHR@5 0.922) does not survive at scale: fusing a single-pass dense encoder that is itself weak on diverse papers, as the retrieval results above show, adds nothing here, which is why BM25-primary is the production default. This metric measures whether retrieval surfaces support for a known claim, not whether the generated answer is faithful; the latter is Section 6.3.

6.3 Generation Faithfulness

We first report the faithfulness of the system's generated answers on the same 51 queries, using the local model gemma4:12b (Gemma Team 2026) for generation. The mean generation-faithfulness score is 0.971: 97.1% of the atomic claims in the generated answers are entailed by the retrieved context. The mean contradiction rate is 0.0, so no answer atom was classified as contradicting its source. Of 200 inline citations checked, 188 are fully supported, 12 partially supported, and none unsupported, a citation-support rate of 0.94. These scores measure grounding to the retrieved context and the absence of outright contradiction, both of which run naturally high for a well-behaved retrieval-augmented system, so we read them conservatively. The zero unsupported-citation count is automated evidence for the indirect citation mechanism, not a claim of perfect correctness against ground truth; assessing that is the job of the human evaluation (Section 7).

6.4 Does the Grounding Hold Across Models?

ScholAR's premise is that grounding lives in the retrieval and application layer, not in the model, so we swapped only the generation model and re-ran the pipeline over the 100 diverse

Table 5: The same ScholAR pipeline with only the generation model changed, on the 100 diverse cases. Faith. is mean generation faithfulness and Hall. the mean hallucination rate; Cites is the number of inline citations the model emitted and Supp. the fraction of them supported by the cited evidence. Faithfulness varies widely across models; citation support does not.

Model	Faith.	Hall.	Cites	Supp.
gemma4:12b	0.947	0.001	289	0.879
qwen3.5:9b	0.888	0.000	579	0.872
mistral:7b	0.819	0.006	205	0.839
llama3.1:8b	0.719	0.046	85	0.941

cases (Table 5). Only half of that premise survives. Generation faithfulness does *not* hold constant: it ranges from 0.719 (llama3.1:8b) to 0.947 (gemma4:12b), and llama3.1:8b carries the only non-trivial hallucination rate (0.046). Citation support, by contrast, stays within a narrow band for every model (0.839 to 0.941). The layer the design actually controls, the evidence-identifier citation mechanism, is the stable one; the prose around the citations tracks the strength of the underlying model.

llama3.1:8b is the instructive case: it emits the fewest citations (85, against 579 for qwen3.5:9b) yet has the highest support rate (0.941). When it cites, it cites well, but it answers tersely and leaves more of its prose uncited, which is what drags its faithfulness down. A high citation-support rate alone is therefore not evidence of a faithful answer, which is why we report both. We read this as refining the model-agnostic claim rather than confirming it: the citation mechanism transfers across models, answer quality still depends on the model.

6.5 Visual Grounding

On the 18-case visual pilot, retrieval routes to the correct figure or table in all 18 cases, with no case falling back to the caption-only path. We also compare two local multimodal models on a keyword-overlap answer-quality proxy (not an LLM-judge): gemma4:12b scores 0.731 and qwen3.5:9b (Qwen Team 2026) scores 0.812. A caption-only ablation, sending the same prompt without the image, scores 0.752 versus 0.731 for full vision under gemma4:12b. We do not read this as evidence that the image is unnecessary: the proxy rewards lexical overlap with the question rather than grounded visual correctness, and many answers are also restated in nearby prose. Separating these effects requires a correctness-based judge, which we leave to the human study.

6.6 Multi-Document Localization

On the 10 arXiv-resolvable cases, ScholAR recovers the correct cited paper in the top 5 half the time (Recall@5 0.50, MRR 0.183) and never at rank 1 (Recall@1 0.00). A random-guess floor over the same candidate pool (on average 8.0 candidate papers per case; floor Recall@5 0.625, MRR 0.340) puts the system at or below chance on this small benchmark.

Table 6: ScholAR versus local baselines on the shared model (qwen3.5:9b) and cases (51 labeled + 40 diverse). Faith. is generation faithfulness; Correct. is must-include recall; Cite-F1 is ALCE-style citation F1. Every emitted page citation is valid (invalid-page rate 0 for all systems).

System	Faith.	Correct.	Cite-F1
PDF-chat (long-context)	0.801	0.267	0.742
Vanilla RAG	0.860	0.340	0.802
PaperQA2-style (RCS, local)	0.872	0.550	0.782
ScholAR	0.892	0.563	0.760

An oracle that restricts retrieval to the anchor plus the single correct paper still reaches only Recall@1 0.00 and MRR 0.289, which points to a difficulty that is not only cross-paper competition but also within-paper chunk ranking on these questions. This is the clearest open problem in the system, and it echoes M3SciQA's finding that even large models struggle at cross-paper localization (Li et al. 2024).

6.7 Comparison with Local Baselines

To place ScholAR against alternatives under its own local constraint, we reimplement three baselines on the same model (qwen3.5:9b) and the same cases (51 labeled and 40 diverse): long-context PDF-chat, which puts the whole paper in the model's context; vanilla RAG, with fixed cross-page chunks and dense retrieval; and a local reimplementation of the rerank-and-summarize step of PaperQA2 (Skarlinski et al. 2024). Table 6 shows ScholAR is strongest on the two axes that matter most, generation faithfulness (0.892) and answer correctness (0.563), narrowly ahead of the PaperQA2-style system and well ahead of long-context PDF-chat, which is worst at answering correctly (0.267): retrieval beats putting the whole document in context. ScholAR is not best on every axis, since the terser baselines reach higher citation precision by citing less, but it leads where grounding and correctness are concerned, with a simpler pipeline than the rerank-and-summarize baseline and page citations that are valid by construction. We compare against local reimplementations, not the hosted OpenScholar, PaperQA2, or SciRAG, which depend on cloud models and large datastores; this is a resource-matched local comparison, not a claim to frontier-scale quality.

6.8 Abstention on Unanswerable Questions

A grounded system should decline when the answer is not present rather than invent one. Every case above is answerable, so we built a complementary set of 20 provably-unanswerable questions: each paper-specific question is posed against a different paper in which the queried fact is provably absent (verified by exact substring over that paper's text), so the correct response is to abstain. Table 7 reports how often each model declines. All four abstain on the large majority of cases: qwen3.5:9b and llama3.1:8b decline on every question, gemma4:12b on 19 of 20, and mistral:7b on 18 of 20. The few non-abstentions are not free invention but a benign overlap: on a question about minibatch size, the wrongly paired paper independently reports a minibatch size, so the model surfaces a real but off-target value. The indirect-evidence prompt, not any one model, drives this behavior. The set is small and synthetic, and abstention is scored by a refusal-phrase detector that we validated by reading every model output, so we read this as evidence that grounding induces abstention rather than as a precise rate.

Table 7: Per-model behavior of the unmodified ScholAR pipeline: only the generation model changes. Abstain / Fabricate are measured on the 20 provably-unanswerable questions (Abstain is the fraction correctly declined). Mem. is the loaded footprint in GB (weights plus KV cache) and latency is end-to-end in seconds (mean / p95) over 20 questions on the evaluation laptop; retrieval adds about 40 ms for every model.

	Unanswerable		Cost per query		
Model	Abst.	Fabr.	Mem.	Tok/s	Lat.
qwen3.5:9b	1.00	0.00	6.1	21.9	11.4/14.6
gemma4:12b	0.95	0.05	8.1	16.3	8.8/14.2
llama3.1:8b	1.00	0.00	6.9	26.9	6.3/17.8
mistral:7b	0.90	0.10	6.5	24.9	6.1/8.4

6.9 Local Deployability

Because ScholAR targets a single local machine, we also measured what a query costs there. On the same laptop (Apple Silicon, 18 GB), we timed the answering path (BM25 retrieval, then generation) over 20 questions for each of the four local models. Retrieval is not the bottleneck: BM25 over one paper's chunks returns in about 40 ms regardless of model, so end-to-end latency is set almost entirely by generation. Every model fits within the 18 GB budget (roughly 6 to 8 GB loaded, weights plus KV cache) and answers in single-digit to low-double-digit seconds (Table 7). The text-only 7 to 8 billion parameter models (mistral:7b, llama3.1:8b) are fastest, at about 6 seconds and 25 tokens per second; the larger multimodal models cost more per token but stay interactive. No cloud call and no GPU cluster: one consumer laptop runs the whole loop.

7 Human Evaluation (In Progress)

Automated metrics measure component behavior; they do not tell us whether a reader would judge an answer correct and well-cited. We therefore designed a model-agnostic human evaluation, grounded in the practice of comparable systems (Asai et al. 2024; Ding et al. 2025; Skarlinski et al. 2024), over the 100-case diverse benchmark of Section 5. Each question is answered by four local generation models (qwen3.5:9b, gemma4:12b, llama3.1:8b (Llama Team 2024), mistral:7b (Jiang et al. 2023)) running the *same* ScholAR pipeline, so only the generation model changes. Expert evaluators score every answer on four anchored 1–5 dimensions, Relevance, Coverage, Faithfulness, and Usefulness, and grade each inline citation as Supported, Partial, or Unsupported. Answers are shown blind and in randomized

Table 8: One question through the unmodified pipeline, verbatim from the logs. The middle row is a real failure we do not hide: the answer is right, but a surplus citation points at a sentence that does not support it, which is what keeps citation precision below 1.0 (Table 6).

Question: What is the size of a minibatch used for each SGD update?
Gold fact: "a minibatch of 80 sentences" (Bahdanau, Cho, and Bengio 2015), p. 5

(i) Asked of the paper that contains the answer
gemma4:12b: "The training process utilized a minibatch size of 80 sentences for calculating each SGD update direction [1]."
[1] → p. 5: "Each SGD update direction is computed using a minibatch of 80 sentences." **supported**

(ii) The same pipeline, a surplus citation
qwen3.5:9b: same correct answer, but additionally cites p. 2: "In this paper, we show that the proposed approach of jointly learning to align and translate. . . " **unsupported**

(iii) Asked of a paper that does not contain the answer
qwen3.5:9b: "The provided evidence does not state the specific minibatch size used for each SGD update, but rather discusses optimal sizes in different contexts." **abstains**

Table 9: Human-evaluation results (in progress). Values are placeholders pending expert scoring; they are not estimated. Dimensions are mean 1–5.

Model	Rel.	Cov.	Faith.	Use.	Cite-F1
qwen3.5:9b	–	–	–	–	–
gemma4:12b	–	–	–	–	–
llama3.1:8b	–	–	–	–	–
mistral:7b	–	–	–	–	–

order. From the scores we compute per-model means and citation precision, recall, and F1; a Friedman test across the four models to assess whether grounding is model-agnostic; and the correlation between human Faithfulness scores and the automated faithfulness metric.

The four models' answers are generated across the diverse benchmark; the expert scoring itself is the remaining step. Table 9 lists the reported quantities as placeholders; the values will be filled from the completed scoring and are not estimated here.

We also report, once available, the Friedman-test result on the Faithfulness dimension (to be filled) and the human-to-automated Faithfulness correlation (to be filled), which together test whether ScholAR's grounding holds across local models and whether the automated metric tracks expert judgment.

8 Discussion and Limitations

ScholAR's contribution is a fully local integration whose two genuinely novel elements, page-preserving chunking and indirect citation grounding, are simple and auditable, and in which the generated answers are measurably grounded, with no contradictions and no unsupported citations on the studied set. The limits of that evidence set the agenda for the rest of this section.

The retrieval and retrieval-support benchmarks now span 100 cases across 25 papers, but their gold chunks are auto-derived from mined facts rather than hand-labeled, and the mined queries carry lexical overlap that likely flatters lexical retrieval; the 3-paper hand-labeled sets remain the higher-precision anchor. The visual (18 cases) and multi-document (10 to 18 cases) benchmarks are still small and hand-labeled on three papers, so several of their deltas fall within noise and carry no significance testing. The abstention probe is likewise small and synthetic (20 auto-constructed cross-paper negatives scored by a validated refusal detector), so its near-total abstention rates indicate a tendency rather than a guarantee. The generation-faithfulness metric rewards grounding to retrieved context and the absence of contradiction, which leaves it insensitive to subtle unfaithfulness, and that gap is exactly what the human evaluation exists to close.

Our comparison (Section 6.7) is against local reimplementations of PDF-chat, vanilla RAG, and PaperQA2's rerank-and-summarize step, not against the hosted OpenScholar, PaperQA2, or SciRAG. Those systems depend on cloud-hosted models and large datastores, so a head-to-head would require either a cloud run, contradicting our local-only design, or crippling them to an untested backend; the honest scope of our result is therefore resource-matched local systems, not frontier-scale quality. Finally, cross-document localization is at or below a random floor and the oracle result suggests a within-paper ranking difficulty on these questions, which is our clearest open problem.

9 Conclusion

We presented ScholAR, a fully local retrieval-augmented system for grounded scientific document comprehension that combines page-preserving chunking with indirect citation grounding and extends to visual and multi-document questions. On 51 cases its generated answers are grounded with a mean score of 0.971, no contradictions, and 94% fully supported citations; retrieval reaches Recall@5 of 0.93 across 25 papers; and figure and table routing is correct in all 18 pilot cases. Against local baselines ScholAR leads on answer correctness and faithfulness, and on unanswerable questions the local models decline rather than fabricate 90 to 100% of the time. Swapping the generation model shows the citation layer transfers while answer faithfulness does not, so grounding is model-agnostic only where the design enforces it. Cross-document localization is at or below chance and is our clearest open problem. We contribute a 100-case, 25-paper benchmark and a human-evaluation protocol whose expert scoring will replace the placeholders reported here.

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